SCHOOL OF ENGINEERING & COMPUTING

PROJECT ISO: TABLETOP CLEANROOM

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BACKGROUND

•Semiconductor chip manufacturing requires extremely clean environments to prevent particle contamination and defects
•ASML collaborates with Fairfield University labs to study cleanroom systems in conjunction with mechatronic automation and system behavior

OBJECTIVES

- ISO 14644-1 Compliance: Achieve ISO 6 (≤ 35,000 particles ≥ 0.5 µm/m³)
- ISO 6 chosen due to budget restrictions and particle count requirements and complexity
- Cost Efficiency: Design a system lower in cost than traditional cleanrooms.
- Automated Mitigation: Integrate automation for hands free movement within cleanroom to prevent contamination
- Real-Time Telemetry: Design a real-time dashboard for continuous monitoring and ISO compliance alerts

METHODOLOGY

- CAD & Mechatronics: Designed a 30” × 20” × 15” enclosure in SolidWorks using 80/20 T-slot framing. Integrated dual linear actuators, a ball screw system, and a servo gripper for automated material transfer
- Power & Control Systems: Utilized an LM2596 buck converter and op-amp circuitry to regulate voltage across the entire system
- CFD Analysis: Used Ansys Fluent to optimize fan and HEPA filter placement, achieving laminar airflow at 0.45 m/s to meet ISO 6 standards.
- Sensor Integration: Deployed SHT31, SDP31, and PMS7003 sensors with an Arduino Uno, streaming environmental data via JSON
- GUI Development: Built a Web Serial API dashboard (HTML/CSS/JS) for real-time visualization, monitoring temperature, humidity, and particle count with automated ISO threshold alerts

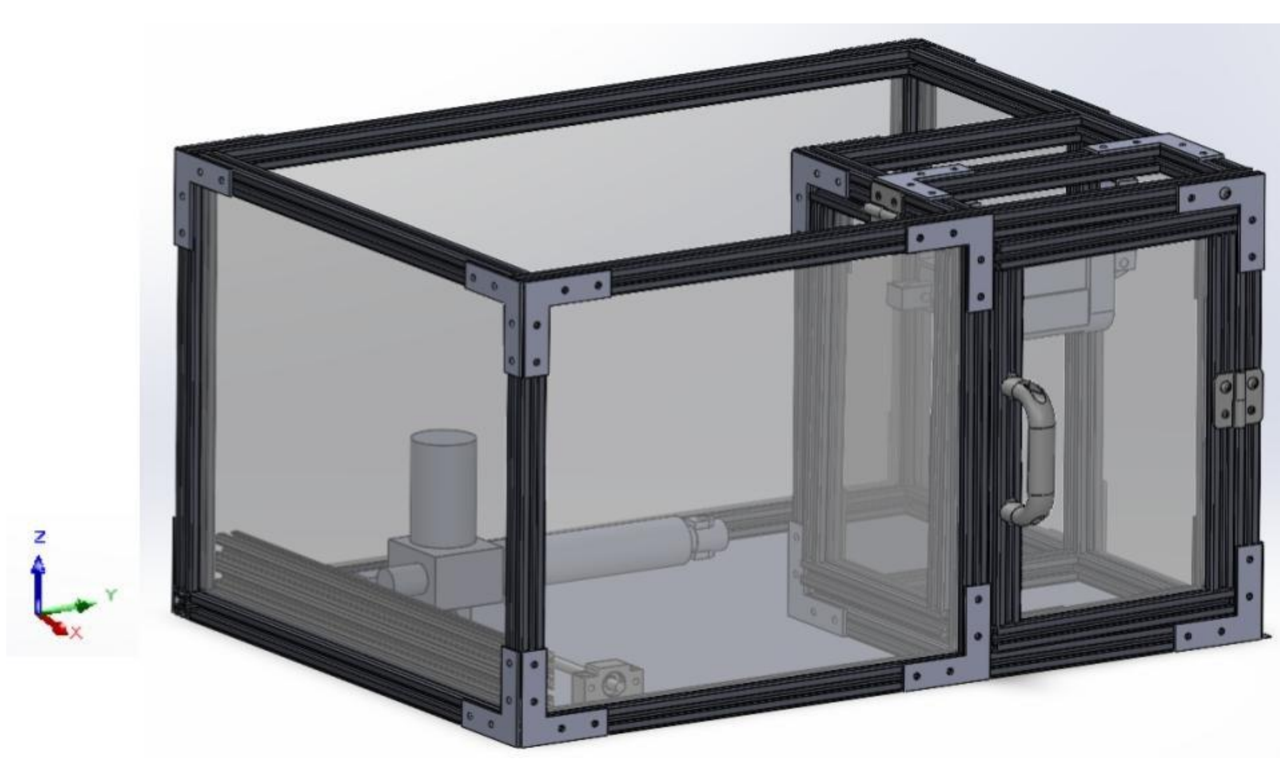

Fig. 1. SolidWorks model of 30” x 20” x 15” tabletop cleanroom. Global coordinates to show actuator and ball screw movement and direction

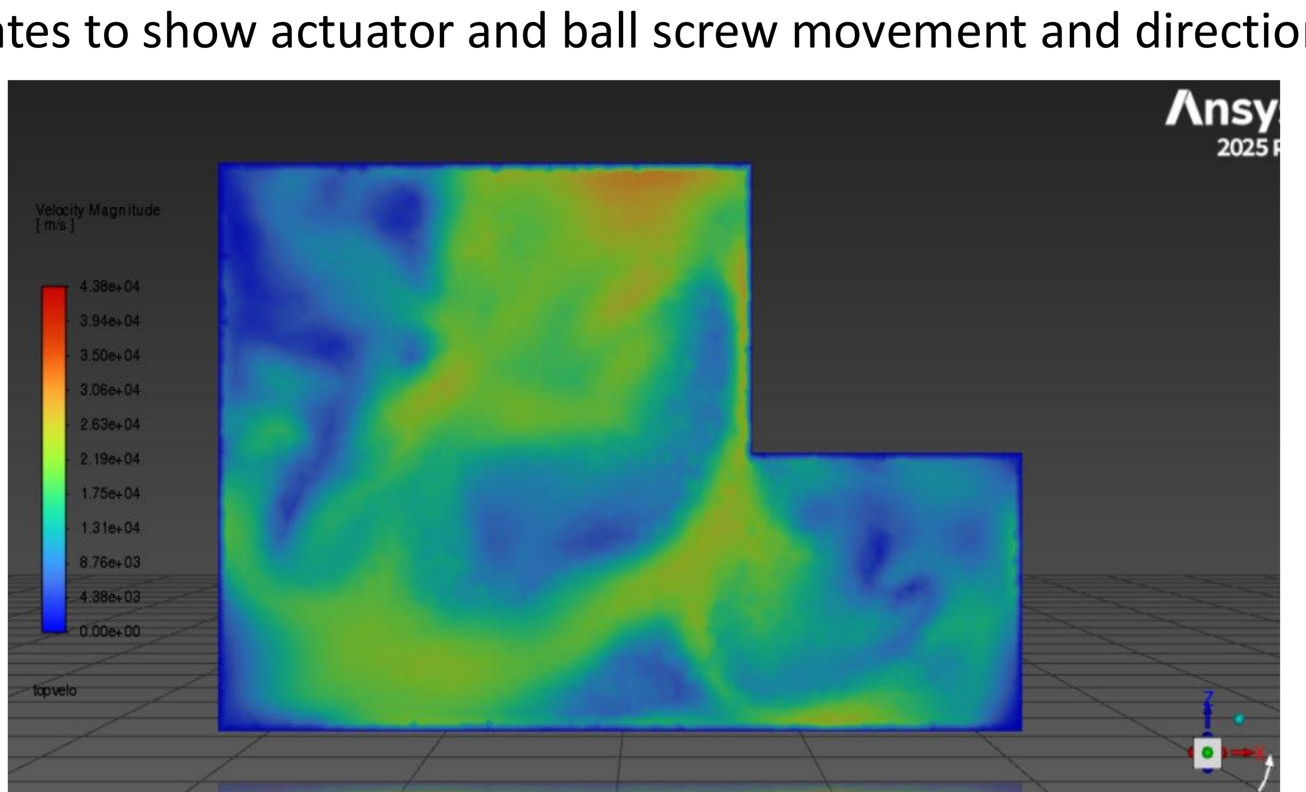

Fig. 2. Ansys CFD cross-sectional model of the cleanroom with 1 inlet and 1 outlet

ISO Class	Maximum Particles/m³						Particles/ft³	FS 209E Equivalent
	≥0.1µm	≥0.2µm	≥0.3µm	≥0.5µm	≥1µm	≥5µm	≥0.5µm	
ISO 6	1,000,000	237,000	102,000	35,200	8,320	293	1,000	Class 1,000

Fig. 3. Particle count for ISO 6 standards, which was referenced for the design of both the cleanroom and airflow system

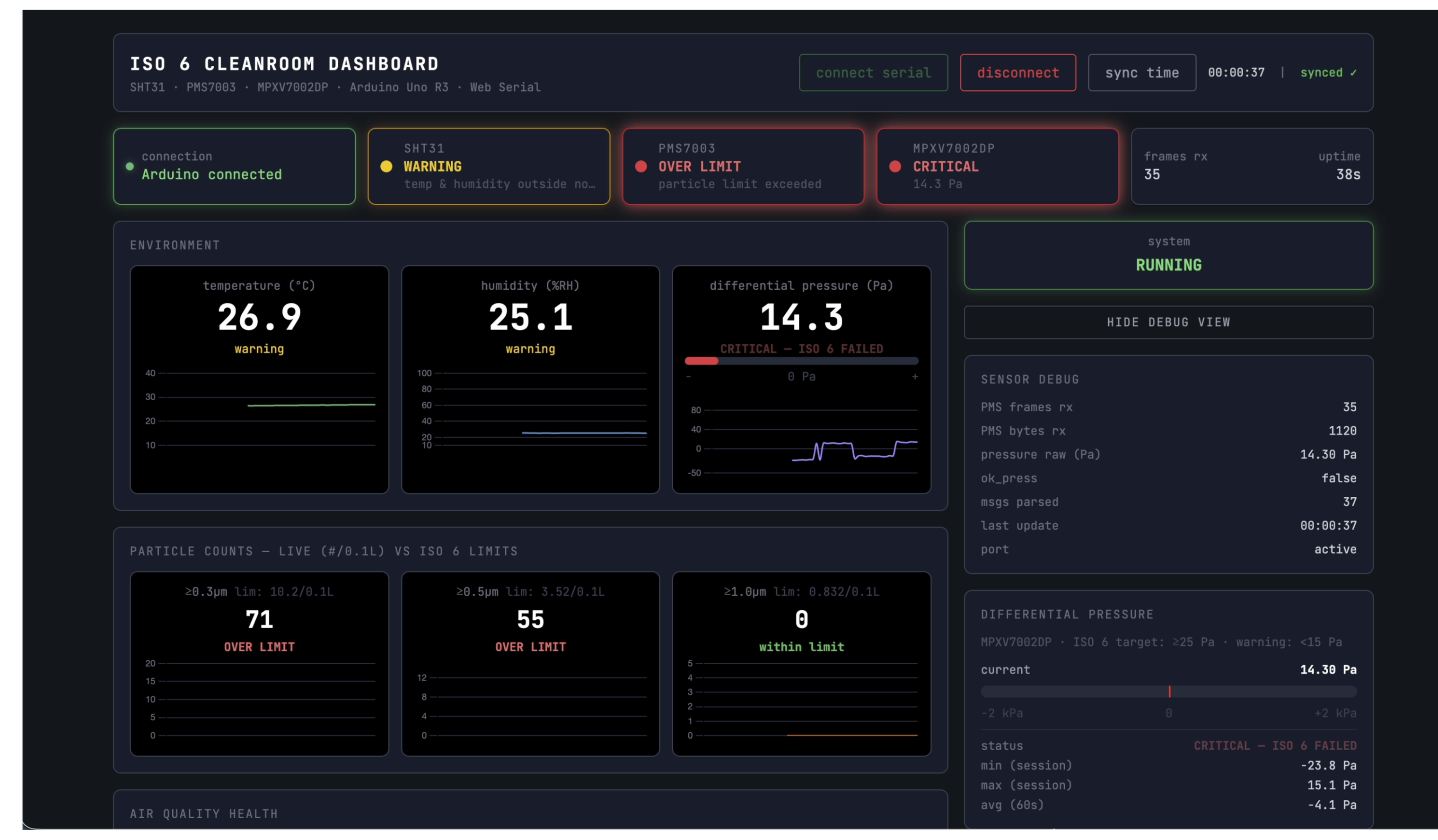

Fig. 4. Browser-based control GUI providing real-time ISO 14644-1 monitoring with color-coded compliance and sensor status indicators

RESULTS

- System Integration: Successfully developed a fully integrated enclosure with mechatronic manipulation for automated material handling
- Airflow Performance: Implemented a filtration system that delivers clean air while exhausting potential particulates from the chamber
- Sensor Validation: Integrated SHT31 (temp/humidity), SDP31 (pressure), and PMS7003(particle count) sensors to measure ISO status integrity
- GUI Functionality: Deployed a web-based interface with low latency for real-time room monitoring and system control via browser

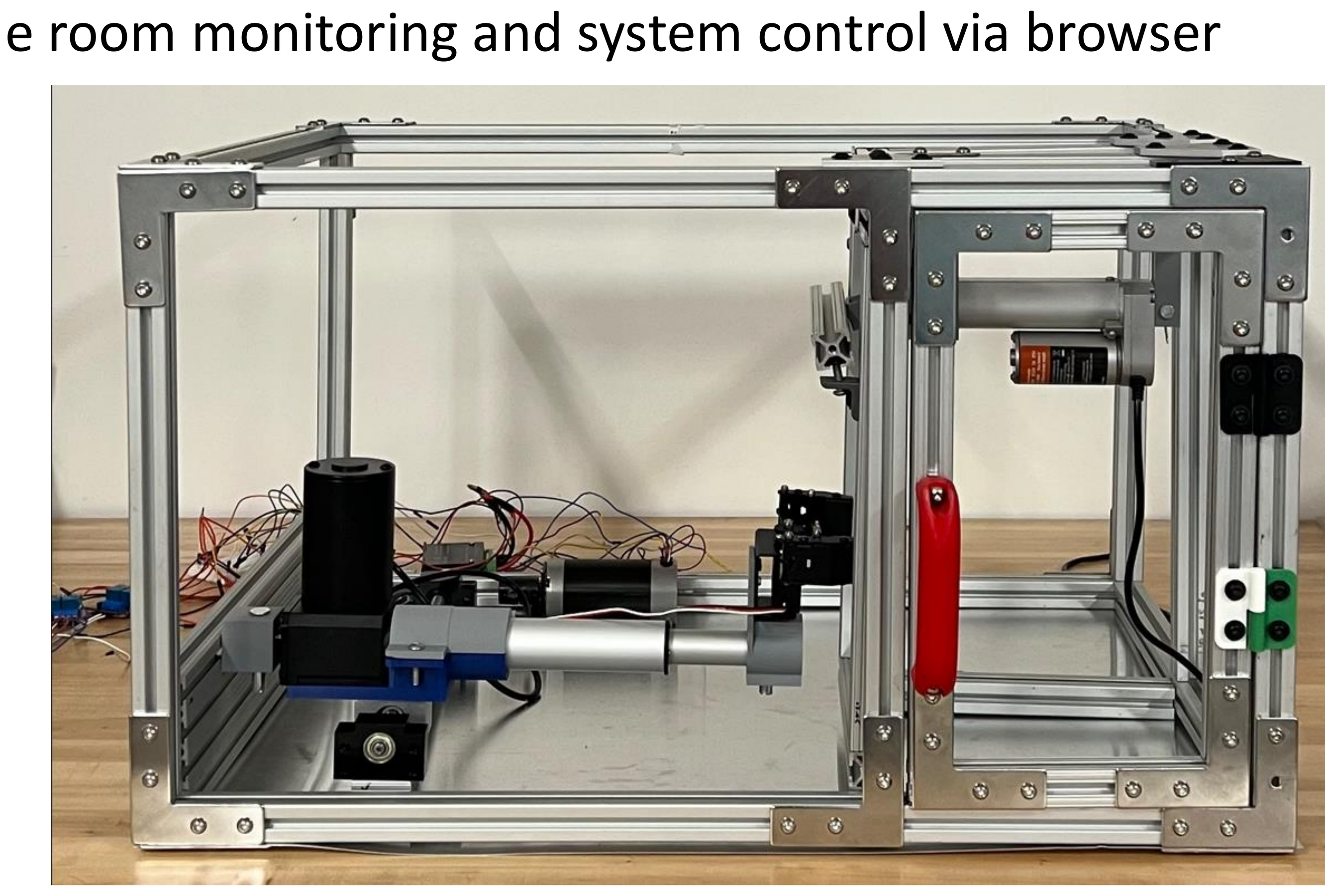

Fig. 5. Tabletop cleanroom enclosure with mechatronics integration including two linear actuators, a gripper servo motor, and a ball screw with a brushless motor. Whole system is controlled via an Arduino Uno R4.

CONCLUSIONS

- Developed a low-cost cleanroom platform for academic research and testing applications
- Demonstrated that web-based telemetry provides a scalable, accessible alternative to closed-source systems

ACKNOWLEDGEMENT

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